Is Schema Complexity One-Dimensional? A Controlled Empirical Study of Type Count, Confusability, and Their Interaction in Training-Free LLM Information Extraction

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Abstract

Training-free LLM information extraction degrades severely on complex schemas, yet what makes a schema "complex" remains poorly understood. We present the first controlled empirical study that disentangles two dimensions of schema complexity—*type count* and *type confusability*—across three IE tasks (NER, event detection, relation extraction), two LLM families (Qwen3-32B, LLaMA-3.3-70B), and two embedding spaces (Qwen3, MiniLM). Our findings reveal that schema complexity is not one-dimensional. (1) Pairwise type confusability significantly predicts type confusion errors on certain model×embedding×task combinations (r=0.156, $p<10^{-11}$ for Qwen3 on Few-NERD; r=0.115, $p<10^{-42}$ for MiniLM on MAVEN), but the signal is highly model-dependent—LLaMA-3.3-70B shows no significant effect (r=0.042, p=0.068). (2) Count and confusability degrade performance through distinguishable error channels: count drives entity omission (miss rate 0.28→0.55), while confusability drives type substitution (4.8× higher in high-confusability batches). (3) Per-type confusability aggregation introduces a centrality confound in semantically homogeneous schemas, where pairwise analysis remains valid. (4) Counter-intuitively, confusability-minimizing schema decomposition *harms* performance (Δ=−3.3 to −7.3 F_1 vs. single-pass), because type co-presence provides disambiguation context that outweighs confusion costs. These findings establish that confusability effects are real but not universal, and that the intuitive "separate confusable types" strategy fails for LLMs that leverage comparative type reasoning.

1 Introduction

Training-free LLM information extraction—using instruction-tuned LLMs with in-context type definitions but no task-specific training—has emerged as a practical paradigm for rapid deployment. However, performance degrades sharply as schema complexity increases: schemas with 50+ types can exhibit 20−60 F_1 gaps compared to supervised baselines.

Prior work has addressed this degradation through schema decomposition: splitting large schemas into smaller batches for independent extraction. UniNER [1] uses per-type splitting, processing one type at a time. GoLLIE [2] provides type definitions as structured code. LLM-PEE [3] implicitly leverages similarity-based grouping. Yet none of these approaches validates the *assumption* that drives decomposition: what exactly makes schemas complex for LLMs, and does reducing a specific complexity dimension actually help?

We pose a precise question: **Is schema complexity one-dimensional, or do distinct dimensions degrade LLM extraction through separable mechanisms?** Concretely, we isolate two candidate dimensions: *type count* (how many types the LLM must consider simultaneously) and *type confusability* (how semantically similar co-present types are). Through controlled experiments across three IE tasks, two LLMs, and two embedding spaces, we find:

1. **Pairwise confusability predicts type confusion**, but the effect is model- and embedding-dependent. Qwen3-32B on Few-NERD shows r=0.156 ($p<10^{-11}$); LLaMA-3.3-70B on the

same task shows r=0.042 (not significant). No single embedding universally captures confusability across tasks.

2. **Count and confusability act through distinguishable channels**: count increases entity omission (recall degrades monotonically), while confusability increases type substitution errors (4.8× ratio in extreme conditions). However, a formal 2×2 interaction test is not significant (p=0.582), consistent with weak additive independence.
3. **Per-type confusability aggregation is confounded** in semantically homogeneous schemas. On MAVEN (168 FrameNet-derived event types), per-type mean confusability correlates *positively* with F_1 (ρ=+0.195), because central, well-represented types have high mean similarity but are also easier to extract. Pairwise analysis avoids this confound.
4. **Decomposition fails**: all confusability-minimizing decomposition strategies (SDGD-zero, SDGD-cal, per-type splitting) perform *worse* than single-pass extraction, because separating confusable types removes the disambiguation context that LLMs use for comparative type reasoning.

These results establish schema complexity as a multi-dimensional phenomenon whose effects are contingent on model architecture, embedding choice, and schema structure. The negative decomposition result is itself informative: it reveals that LLMs exploit type co-presence for disambiguation, a mechanism that simple separation strategies destroy.

2 Related Work

Training-free LLM IE. Recent work has shown that instruction-tuned LLMs can perform IE tasks zero-shot by providing type definitions in the prompt [1, 2, 3]. Performance is competitive on simple schemas (5−15 types) but degrades substantially on complex ones.

Schema decomposition. UniNER [1] processes one type at a time, achieving 3.3 F_1 gains on some benchmarks but at $k\times$ API cost. LLM-PEE [3] groups types by embedding similarity, implicitly targeting confusability reduction. Neither approach validates whether the targeted complexity dimension is actually causal.

Schema complexity analysis. The relationship between label set complexity and extraction performance has been studied in supervised settings [4, 5], but controlled isolation of individual complexity dimensions in the LLM setting is unexplored. Our work fills this gap with factorial experimental designs.

3 Experimental Setup

3.1 Tasks and Datasets

We study three IE tasks with varying schema characteristics:

- **Few-NERD** (NER): 66 fine-grained entity types across 8 coarse domains. Cross-domain schema with heterogeneous type semantics.
- **MAVEN** (Event Detection): 168 event types derived from FrameNet. Same-domain schema with homogeneous type semantics.
- **DocRED** (Relation Extraction): 96 relation types. Document-level task where extraction involves cross-sentence reasoning.

3.2 Models

- **Qwen3-32B-Instruct**: Primary extraction model. All experiments except the cross-model study use this model.
- **LLaMA-3.3-70B-Instruct**: Cross-model validation (Section 4.3).

3.3 Confusability Measures

We compute *Schema Confusability Score* (SCS) as the cosine similarity between type definition embeddings:

$$\text{SCS}(t_i, t_j) = \cos(\mathbf{e}(t_i), \mathbf{e}(t_j)) \tag{1}$$

where $\mathbf{e}(\cdot)$ denotes the embedding function. We use two embedding models:

- **Qwen3 embeddings**: Derived from the extraction model's own representations.
- **MiniLM**: A general-purpose sentence embedding model (all-MiniLM-L6-v2).

Pairwise confusability uses SCS directly between type pairs. *Per-type confusability* (Batch Confusability Score, BCS) averages SCS across all co-present types: $\text{BCS}(t_i) = \frac{1}{|B|-1} \sum_{t_j \in B \setminus \{t_i\}} \text{SCS}(t_i, t_j)$.

3.4 Evaluation Protocol

For NER and event detection, we report both relaxed F_1 (partial span matching) and exact F_1. We use controlled batch construction: for each *count* level, we sample random subsets of types and evaluate extraction on held-out test data. Within-count experiments vary confusability (BCS) while holding count fixed, isolating the confusability effect.

4 Finding 1: Pairwise Confusability Predicts Type Confusion

We investigate whether pairwise type confusability (SCS between type pairs) predicts actual type confusion errors. We test this across tasks, models, and embeddings.

4.1 Few-NERD: Qwen3 Embeddings

On Few-NERD with Qwen3-32B extraction and Qwen3 embeddings, pairwise SCS significantly correlates with pairwise confusion rate: Pearson r=0.156 (p=4.6×10^{-12}, n=1,953 type pairs). Among type pairs with nonzero confusion, Spearman ρ=0.254 (p=1.1×10^{-6}). High-SCS pairs such as `person-other`↔`person-politician` (SCS=0.934) account for the majority of type confusion errors.

In extreme batches (deliberately constructed to maximize or minimize BCS), the confusion rate ratio reaches 10−28× between high- and low-confusability conditions. Random batches show a smaller but consistent ratio of 1.7−2.4×.

4.2 MAVEN: Embedding-Dependent Signal

MAVEN presents a contrasting pattern across embeddings (Table 5). With MiniLM embeddings, the pairwise signal is significant: r=0.115 (p=2.1×10^{-42}, n=14,028 pairs), with Spearman ρ=0.128 (nonzero-only ρ=0.185). However, Qwen3 embeddings yield r=−0.009 (p=0.38)—completely non-significant.

This divergence reveals that confusability measurement is not embedding-agnostic. In semantically homogeneous schemas (MAVEN's FrameNet-derived types), the extraction model's own embedding space may not differentiate confusable type pairs effectively, while a general-purpose embedding preserves discriminative structure.

Table 1: Cross-model comparison on Few-NERD. Confusability effects are substantially weaker for LLaMA-3.3-70B across all metrics.

Metric	Qwen3-32B	LLaMA-3.3-70B
Baseline F_1 (count 64)	0.414	0.378
Pairwise r (Qwen3 embed)	0.156***	0.042 (ns)
Per-type ρ	-0.293*	-0.082 (ns)
Extreme confusion ratio	10–28×	2.1–2.5×
Random confusion ratio	1.7–2.4×	1.15–1.2×

Table 2: LLaMA-3.3-70B detailed statistics on Few-NERD. No analysis reaches significance.

Analysis	Statistic	Value	p	Verdict
Pairwise	Pearson r	0.042	0.068	NS
Pairwise (nonzero)	Spearman ρ	0.078	0.120	NS
Per-type	Spearman ρ	-0.082	0.538	NS
Per-type (partial)	Partial r	-0.055	0.684	NS

4.3 Cross-Model Validation: LLaMA-3.3-70B

To test whether confusability effects generalize across model families, we replicate the Few-NERD protocol with LLaMA-3.3-70B-Instruct. Table 1 presents the comparison.

LLaMA-3.3-70B shows no significant pairwise signal (r=0.042, p=0.068), no significant per-type correlation (ρ=-0.082, p=0.538), and dramatically lower confusion rate ratios (2.1$-$2.5× vs. 10$-$28×). Even controlling for log-frequency, the partial correlation remains non-significant (partial r=-0.055, p=0.684).

This 4× effect size difference (r=0.156 vs. r=0.042) across model families implies that confusability sensitivity is model-dependent. Prior conflicting findings about schema complexity effects may be partly explained by this model dependence.

4.4 Embedding Proxy Validation

To assess whether embedding-based SCS is a valid proxy for actual model confusion, we compare SCS rankings against an empirical confusion matrix derived from calibration data (Table 3).

Qwen3 SCS achieves ρ=0.125 (p=0.031) against empirical confusion—statistically significant but explaining only ∼1.6% of confusion variance. The false positive rate of 0.42 indicates that 42% of type pairs predicted as confusable by SCS are not actually confused by the model. Confusability is mediated by factors beyond semantic similarity, including surface form similarity, type definition wording, and in-context position.

4.5 Summary: Pairwise Signal Matrix

Table 5 summarizes the full cross-task × cross-model × cross-embedding matrix.

5 Finding 2: Count and Confusability Degrade Through Distinguishable Channels

5.1 Count Curves Across Tasks

We construct count curves by varying the number of types presented to the LLM while holding other factors constant. The three tasks show strikingly different patterns:

Table 3: Correlation between embedding SCS and empirical confusion rates. Both embeddings show weak predictive power.

Embedding	**Spearman** ρ	p	**Kendall** τ	p
Qwen3	0.125	0.031	0.092	0.044
MiniLM	0.098	0.082	0.071	0.115

Table 4: SCS proxy error analysis: false positive/negative rates for confusion prediction.

Metric	**Qwen3**	**MiniLM**
False Positive Rate	0.42	0.48
False Negative Rate	0.35	0.40
Top-1 Match	Yes	No

Few-NERD. Relaxed F_1 decreases monotonically with count (previously established), with recall declining as the primary driver.

MAVEN. Table 6 shows the *opposite* pattern: relaxed F_1 *increases* with count, from 0.098 (count 1) to 0.243 (count 100). Recall increases monotonically (0.07→0.19), while precision is roughly flat (0.15−0.40). This reversal occurs because MAVEN's 168 FrameNet-derived types form a coherent ontology; more types in the prompt provide richer event taxonomy context that helps the model locate each type's position in the hierarchy.

DocRED. Table 7 shows an approximately flat count curve with no clear monotonic pattern. F_1 ranges from 0.185 (count 4) to 0.208 (count 48), with a slight decline at the full schema (count 96, F_1=0.195).

The three-task comparison reveals that count effects are task-dependent: monotonically negative (Few-NERD), monotonically positive (MAVEN), or negligible (DocRED).

5.2 Within-Count BCS–F_1 Correlation

To isolate confusability from count, we examine the correlation between batch-level BCS and F_1 *within* fixed count levels.

MAVEN. Table 8 shows no significant within-count BCS–F_1 correlation, with both individual count levels and the pooled analysis failing to reach significance.

The positive direction (higher BCS → higher F_1) is opposite to the confusability hypothesis and consistent with centrality confounding (Section 6).

DocRED. Within-count BCS–F_1 correlations are similarly non-significant: count 8 (r=0.032, p=0.62), count 16 (r=−0.058, p=0.48), pooled (r=−0.015, p=0.84). The per-type BCS–F_1 correlation is also null (ρ=−0.04, p=0.72).

5.3 2×2 Factorial Design: Count × Confusability Interaction

We test the interaction between count and confusability using a 2×2 factorial design on Few-NERD: {low count=6, high count=22} × {low BCS, high BCS}, with 12 batches per cell (n=48 total).

The count main effect is significant (F=7.82, p=0.008, η^2=0.14), while BCS is marginal (F=2.93, p=0.094, η^2=0.06). The interaction is not significant (F=0.31, p=0.582, η^2=0.006), consistent with an additive (rather than multiplicative) combination of the two channels. Simple

Table 5: Pairwise SCS–confusion signal across tasks, models, and embeddings. Only 3 of 7 tested cells reach significance; no single embedding or model universally works.

Model	Embedding	Few-NERD	MAVEN	DocRED
Qwen3	Qwen3	r=0.156***	r=−0.009	—
Qwen3	MiniLM	—	r=0.115***	—
LLaMA	Qwen3	r=0.042	—	—

*** $p < 0.001$. Cells marked "—" were not tested.

Table 6: MAVEN count curve (500 samples, relaxed matching). Performance *increases* with type count—opposite of Few-NERD.

Count	Relaxed F_1	Exact F_1	Recall	Precision	Pred/Gold
1	0.098	0.008	0.070	0.162	0.43
6	0.112	0.005	0.085	0.165	0.52
12	0.155	0.011	0.118	0.226	0.52
22	0.192	0.016	0.148	0.274	0.54
44	0.238	0.021	0.180	0.350	0.51
64	0.241	0.018	0.182	0.358	0.51
100	0.243	—	—	—	—

effects analysis shows that BCS effects are directionally consistent at both count levels but neither reaches significance individually (low count: Δ=−0.036, p=0.22; high count: Δ=−0.043, p=0.18).

5.4 Error Taxonomy: Mechanistic Evidence for Distinct Channels

We perform a systematic error classification on Few-NERD extraction outputs, categorizing false positives into four types: type substitution, boundary error, spurious extraction, and missed entity.

Type substitution by confusability. In extreme high-BCS batches, the type substitution rate reaches 0.058, compared to 0.012 in extreme low-BCS batches—a 4.8× ratio (Table 12). However, in *random* batches, the ratio drops to 1.5× (0.022 vs. 0.015), indicating that the effect is pronounced only under deliberately extreme conditions.

Missed entities by count. The miss rate increases monotonically with count: 0.28 (count 6) → 0.42 (count 22) → 0.55 (count 64). Crucially, the miss rate does *not* vary with BCS (high BCS: 0.40; low BCS: 0.38; ratio 1.05), confirming that count and confusability affect different error modes.

Boundary errors and hallucination. Neither boundary errors (high BCS: 0.041; low BCS: 0.038; ratio 1.08) nor hallucination (rate 0.03, no BCS effect) are influenced by confusability, as expected—these error modes are orthogonal to type semantics.

6 Finding 3: Per-Type Aggregation Introduces Centrality Confound

6.1 MAVEN: Positive Per-Type Correlation

On MAVEN, per-type mean BCS correlates *positively* with per-type F_1 under both embeddings (Table 14). This is the *opposite* of what the confusability hypothesis predicts.

Table 7: DocRED count curve (document-level RE). Performance is near-flat—schema complexity effects are dominated by cross-sentence reasoning demands.

Count	F_1	Recall	Precision	Confusion Rate
4	0.185	0.145	0.256	0.008
8	0.192	0.152	0.261	0.012
16	0.198	0.158	0.264	0.015
32	0.203	0.165	0.258	0.018
48	0.208	0.170	0.261	0.019
96	0.195	0.160	0.250	0.022

Table 8: MAVEN within-count BCS–F_1 correlation. No significant relationship detected at any count level.

Count	r	p	Direction	Verdict
6	+0.175	0.185	Positive	Fail
12	+0.124	0.315	Positive	Fail
Pooled	+0.142	0.108	Positive	Fail

The mechanism is *centrality confounding*: in MAVEN's semantically homogeneous schema, types with high mean BCS are central, frequent types (e.g., ATTACK, MOVEMENT) that the LLM extracts well because they are well-represented in training data and have clear definitions. Mean BCS conflates confusability with type centrality; per-type aggregation cannot separate these effects.

Crucially, the pairwise analysis tells a different story. MiniLM pairwise SCS significantly predicts confusion (r=0.115, $p<10^{-42}$), demonstrating that pairwise analysis avoids the centrality confound. The Qwen3 pairwise signal is null (r=−0.009), highlighting that embedding choice interacts with schema structure.

The confusion rate analysis further supports this: MAVEN's extreme confusion ratios (5.8× at count 6, 4.2× at count 12) are lower than Few-NERD's (10−28×), while random ratios (1.4×, 1.3×) are modest.

6.2 DocRED: Null Result as Boundary Condition

DocRED yields complete null results across all confusability analyses (Section 5.2). The pairwise signal is non-significant for both Qwen3 (r=0.023, p=0.14) and MiniLM (r=0.031, p=0.08). The per-type BCS–F_1 correlation is null (ρ=−0.04, p=0.72).

This null result establishes a boundary condition: confusability effects require that type definitions occupy a significant fraction of the prompt. In document-level RE, the document content dominates the prompt; type definitions are a small fraction, and performance bottlenecks lie in cross-sentence reasoning and entity pair combinatorics rather than type confusability.

7 Finding 4: Confusability-Minimizing Decomposition Fails

Motivated by the confusability signal in Section 4, we tested whether separating confusable types into different batches (Schema DAG Decomposition, SDGD) could improve extraction. We report a comprehensive negative result: all decomposition strategies perform worse than single-pass extraction. Full details and per-batch breakdowns are in Appendix A.

7.1 Overview of Methods

- **Single-pass**: All types in one prompt.

Table 9: 2×2 cell means (F_1, text match) on Few-NERD. Both factors show directional effects, but the interaction is not significant.

	Low BCS	**High BCS**	**BCS Effect**
Low Count (6)	0.348	0.312	Δ=−0.036
High Count (22)	0.298	0.255	Δ=−0.043
Count Effect	Δ=−0.050	Δ=−0.057	

Table 10: Two-way ANOVA results. Count is significant; BCS is marginal; the interaction is not significant.

Source	F	p	η^2	**Verdict**
Count	7.82	0.008	0.14	Significant
BCS	2.93	0.094	0.06	Marginal
Count×BCS	0.31	0.582	0.006	NS

- **SDGD-zero**: Graph coloring on the SCS-thresholded pairwise similarity graph (no calibration data).
- **SDGD-cal**: Graph coloring on an empirical confusion matrix from 8–10 calibration samples.
- **Random (k)**: Random batching with the same number of batches as SDGD.
- **Per-type**: One type per batch (UniNER-style).

7.2 Results

Table 16 presents the full comparison.

The consistent pattern across both tasks: decomposition increases recall (each batch focuses on fewer types, enabling more aggressive extraction) but devastates precision (without the full type set, the model cannot disambiguate, leading to massive over-extraction). The precision–recall tradeoff is unfavorable: precision losses far exceed recall gains.

SDGD vs. random is not significant. SDGD-zero does not outperform random batching on either task (Few-NERD: Δ=−0.7pp, p=0.68; MAVEN: Δ=−1.2pp, p=0.42). Confusability-aware grouping provides no value over random partitioning.

Per-type splitting is worst. The extreme decomposition—one type per batch—yields the lowest F_1 (Few-NERD: 0.330; MAVEN: 0.108) at the highest cost (66−168× API calls). This confirms that decomposition granularity is inversely related to performance.

Root cause. Decomposition removes *disambiguation context*: when confusable types are separated, each batch sees only one side of a confusable pair, and the model assigns the available type to any plausible mention. Upon merging batches, the same span may receive conflicting type assignments (8% cross-batch span overlap), and the overall prediction count far exceeds gold annotations (pred/gold ratio 1.36 vs. single-pass 0.47).

8 Discussion

Schema complexity is multi-dimensional but context-dependent. Our results establish that count and confusability are distinguishable dimensions of schema complexity, but their

Table 11: Overall error distribution. Spurious extraction dominates; type substitution (the channel most related to confusability) accounts for only 18%.

Error Type	**Proportion**	**Primary Driver**
Spurious Extraction	45%	Prompt format
Missed Entity	25%	Count (recall)
Type Substitution	18%	BCS (confusability)
Boundary Error	12%	Span detection

Table 12: Type substitution rate by BCS condition. The extreme ratio (4.8×) substantially exceeds the random ratio (1.5×).

Condition	**High BCS**	**Low BCS**	**Ratio**
Extreme Batch	0.058	0.012	4.8×
Random Batch	0.022	0.015	1.5×

effects are contingent on task structure, model architecture, and embedding choice. On Few-NERD (cross-domain NER), count dominates; on MAVEN (same-domain ED), more types *help*; on DocRED (document-level RE), neither dimension matters.

Confusability effects are real but not universal. The pairwise confusability signal is significant for Qwen3 on Few-NERD and MiniLM on MAVEN, but absent for LLaMA-3.3-70B and for Qwen3 embeddings on MAVEN. This model dependence suggests that confusability sensitivity is a property of the model's internal representation, not solely of the schema itself. The 4× effect size difference between Qwen3 (r=0.156) and LLaMA (r=0.042) is itself a finding that may explain prior conflicting reports about schema complexity effects.

Per-type aggregation is unreliable. The centrality confound on MAVEN demonstrates that mean BCS (per-type averaged confusability) is not a valid measure in semantically homogeneous schemas. Future work on schema complexity should use pairwise analysis to avoid this confound, or use empirical confusion matrices that bypass embedding proxy limitations.

Decomposition harms LLM IE. The most counter-intuitive finding is that confusability-minimizing decomposition *hurts* performance. LLMs use type co-presence for comparative reasoning: seeing confusable types together enables the model to reason about their distinctions. Removing this context forces the model into uninformed guessing. This insight has direct implications for decomposition-based approaches: they should be designed to *preserve* disambiguation context rather than minimize confusability.

Implications for future work. (1) Confusability-based methods should be evaluated per-model, not assumed universal. (2) Schema decomposition for LLM IE needs fundamentally different principles—perhaps preserving confusable types together while reducing only the non-informative count overhead. (3) Embedding-based confusability proxies need task-specific calibration, as no single embedding works universally.

Limitations. Our study uses two LLM families; more models are needed to characterize the distribution of confusability sensitivity. The 2×2 factorial design has limited statistical power (n=48); larger-scale factorial experiments could detect smaller interaction effects. DocRED results may partly reflect the difficulty of zero-shot document-level RE rather than a true absence of confusability effects.

Table 13: Missed entity rate by count. Monotonically increasing, independent of BCS.

Count	Miss Rate	Δ vs. Count 6
6	0.28	—
22	0.42	+0.14
64	0.55	+0.27

Table 14: MAVEN pairwise and per-type confusability analysis. Per-type correlations are positive (confounded); pairwise MiniLM is significant.

Embedding	Pairwise r (p)	Per-type ρ (p)
Qwen3	-0.009 (0.38)	$+0.195$ (0.018)
MiniLM	$+0.115$ (2.1×10^{-42})	$+0.212$ (0.009)

9 Conclusion

We presented the first controlled empirical study isolating type count and type confusability as dimensions of schema complexity in training-free LLM information extraction. Our experiments across three IE tasks, two LLMs, and two embedding spaces reveal that: (1) pairwise type confusability is a valid predictor of type confusion errors, but only for specific model×embedding combinations; (2) count and confusability degrade performance through distinguishable error channels; (3) per-type confusability aggregation introduces centrality confounds in homogeneous schemas; and (4) confusability-minimizing decomposition is counter-productive because LLMs leverage type co-presence for disambiguation. These findings reframe schema complexity as a multi-dimensional, model-dependent phenomenon and provide methodological guidance for future research on LLM IE with complex schemas.

A SDGD Decomposition: Detailed Results

This appendix provides detailed results for all decomposition experiments, which are reported as negative results in the main paper (Section 7).

A.1 SDGD-zero on Few-NERD

Graph coloring on the Qwen3 SCS-thresholded graph (τ=0.847) produces 8 batches (avg. size 8.25, range 4−12).

Graph statistics: 142 edges, edge density 0.066, chromatic number 8, max intra-batch BCS 0.842. Cross-batch span overlap: 8% of predictions appear in multiple batches. Total predictions: 4,850 (after dedup: 4,120), averaging 8.2 predictions per sample vs. 5.9 gold entities.

A.2 SDGD-zero on MAVEN

On MAVEN (168 types), adaptive threshold τ=0.319 produces 15 batches (avg. size 11.2, range 5−18), with edge density 0.132 and 1,850 edges.

The larger performance gap on MAVEN (−7.3pp vs. −4.7pp on Few-NERD) is consistent with MAVEN's positive count curve: decomposition directly counteracts the benefit of full ontology visibility.

A.3 SDGD-cal on Few-NERD

SDGD-cal uses 8 calibration samples to build an empirical confusion matrix, yielding 85 nonzero confusion pairs. The empirical confusion matrix correlates weakly with embedding SCS (ρ=0.125,

Table 15: DocRED pairwise confusability analysis. No significant signal detected with either embedding.

Embedding	**Pearson** r	p	n **pairs**	**Verdict**
Qwen3	0.023	0.14	4,560	NS
MiniLM	0.031	0.08	4,560	NS

Table 16: Full method comparison. All decomposition methods are inferior to single-pass. Ordering: single-pass > SDGD-cal $\gtrsim$ random $\gtrsim$ SDGD-zero > per-type. Methods are sorted by Few-NERD F_1.

Method	**Few-NERD**			**MAVEN**	
	F_1	Prec.	Recall	Rel. F_1	API
Single-pass	0.414	0.648	0.304	0.241	1×
SDGD-cal	0.381	0.342	0.430	0.188	7–12×
Random	0.374	0.335	0.425	0.180	8–12×
SDGD-zero	0.367	0.318	0.434	0.168	8–15×
Per-type	0.330	0.225	0.620	0.108	66–168×

p=0.031). Graph coloring produces 7 batches (avg. size 9.4).

SDGD-cal marginally outperforms SDGD-zero (+1.4pp) and random (+0.7pp, p=0.58), but the improvement over random is not significant. Even with empirical confusion data, the decomposition penalty persists.

A.4 SDGD-cal on MAVEN

On MAVEN, 10 calibration samples produce 210 nonzero confusion pairs. Graph coloring yields 12 batches (avg. size 14.0).

SDGD-cal outperforms SDGD-zero (+2.0pp) but not random (+0.8pp, p=0.62). The same pattern as Few-NERD: calibration helps marginally, but decomposition remains inferior to single-pass.

A.5 Random Batching Ablation

To isolate the contribution of confusability-aware grouping from the batch count reduction effect, we compare SDGD-zero against random batching with the same number of batches.

On both tasks, SDGD-zero does not significantly outperform random batching, confirming that confusability-aware grouping provides no measurable benefit. SDGD-zero may actually perform slightly *worse* than random, because deliberately separating confusable types removes same-cluster disambiguation anchors that random batching occasionally preserves.

A.6 Per-Type Splitting Ablation

Per-type splitting (one type per batch) represents the extreme of decomposition—maximum confusability elimination but also maximum loss of disambiguation context.

The method ordering is consistent across both tasks: single-pass > SDGD-cal $\gtrsim$ random $\gtrsim$ SDGD-zero > per-type. Decomposition granularity is inversely correlated with F_1: larger batches (more type context) consistently outperform smaller ones. Per-type splitting achieves the highest recall (0.620 on Few-NERD) through aggressive per-type extraction, but precision collapses (0.225) as each batch drastically over-extracts.

Table 17: SDGD-zero vs. single-pass on Few-NERD (66 types).

Method	**F_1**	**Precision**	**Recall**	**Pred/Gold**	**API**
Single-pass	0.414	0.648	0.304	0.47	1×
SDGD-zero	0.367	0.318	0.434	1.36	8×
Δ	−4.7pp	−33.0pp	+13.0pp	+0.89	+7×

Table 18: SDGD-zero per-batch breakdown (top 4 batches, Few-NERD).

Batch	**Types**	**Avg BCS**	**F_1**	**Recall**	**Precision**
0	12	0.812	0.42	0.48	0.38
1	10	0.795	0.38	0.45	0.33
2	9	0.780	0.35	0.42	0.30
3	8	0.763	0.32	0.40	0.27

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Table 19: SDGD-zero vs. single-pass on MAVEN (168 types, relaxed matching).

Method	Relaxed F_1	Exact F_1	API	Wall-Clock
Single-pass	0.241	0.018	1×	1×
SDGD-zero	0.168	0.012	15×	12.8×
Δ	−7.3pp	−0.6pp	+14×	+11.8×

Table 20: SDGD-cal vs. baselines on Few-NERD.

Method	F_1	Δ vs. SP	Precision	Recall	API
Single-pass	0.414	—	0.648	0.304	1×
Random (k=8)	0.374	−4.0	0.335	0.425	8×
SDGD-zero	0.367	−4.7	0.318	0.434	8×
SDGD-cal	0.381	−3.3	0.342	0.430	7× + 8 cal.

Table 21: Full comparison on MAVEN (168 types, relaxed matching).

Method	Relaxed F_1	Δ vs. SP	API
Single-pass	0.241	—	1×
Random (k=12)	0.180	−6.1	12×
SDGD-zero	0.168	−7.3	15×
SDGD-cal	0.188	−5.3	12× + 10 cal.

Table 22: Random vs. SDGD-zero on Few-NERD (k=8, 10 random seeds).

Method	Mean F_1	Std	Range	vs. SP
Random×10	0.374	0.018	[0.348, 0.402]	−4.0pp
SDGD-zero	0.367	—	—	−4.7pp
Δ (SDGD−Random)	−0.7pp, p=0.68 (NS)			

Table 23: Random vs. SDGD-zero on MAVEN.

Method	Relaxed F_1	vs. SP	p (paired)
Random×5	0.180	−6.1pp	—
SDGD-zero	0.168	−7.3pp	—
Δ	−1.2pp, p=0.42 (NS)		

Table 24: Per-type splitting vs. all methods on Few-NERD.

Method	F_1	Precision	Recall	API	Wall-Clock
Single-pass	0.414	0.648	0.304	1×	1×
SDGD-cal	0.381	0.342	0.430	7×	5.5×
Random (k=8)	0.374	0.335	0.425	8×	6.5×
SDGD-zero	0.367	0.318	0.434	8×	6.5×
Per-type	0.330	0.225	0.620	66×	48×

Table 25: Per-type splitting vs. all methods on MAVEN (relaxed matching).

Method	Relaxed F_1	API	Wall-Clock
Single-pass	0.241	1×	1×
SDGD-cal	0.188	12×	10×
Random	0.180	12×	10×
SDGD-zero	0.168	15×	12.8×
Per-type	0.108	168×	135×